

USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester MCA (CBCS) Degree Examinations
 December 2024

22MCA351 – CLOUD COMPUTING AND BIG DATA ANALYTICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are the key techniques employed in uncovering hidden patterns within unstructured data? Explain.	10	L*1	1	1
	b) "It is easy to work with structured data", Give reasons.	10	L2	1	1
2.	a) How is structured data generated? Explain the sources that generates structured data?	10	L4	1	1
	b) "Explain typical data warehouse and Hadoop environment coexistence with diagram?"	10	L2	1	1
Unit – II					
3.	a) Illustrate the CAP theorem with a real-world example.	10	L4	2	1
	b) Write a note on the following.				
	i) Greatest challenges that prevent business from capitalizing on big data	10	L4	2	1
	ii) Explain Shared nothing architecture	10	L4	2	1
4.	a) Why NOSQL is used? What are the advantages of NOSQL?	10	L3	2	1
	b) How can big data analytics be categorized? Compare analytics 1.0, analytics 2.0, analytics 3.0	10	L1	2	1
Unit – III					
5.	a) With a neat architecture diagram, explain HDFS.	15	L2	3	1
	b) Explain the data management technique of Hadoop?	5	L2	3	1
6.	a) What is Hadoop? Compare SQL with Hadoop?	10	L2	3	1
	b) Explain the components of Hadoop ecosystem?	10	L2	3	1
Unit-IV					
7.	a) Compare and contrast traditional and cloud computing.	10	L4	4	1
	b) What are the building blocks of a private cloud, and how do they work together to provide cloud services?	10	L1	4	1
8.	a) What is cloud deployment? Describe the different cloud deployment models.	10	L2	4	1

P.T.O.

- | | | | | |
|--|----|----|---|---|
| b) Explain the concept of a private cloud and how it differs from a public cloud? What are the key advantages and disadvantages? | 10 | L2 | 4 | 1 |
|--|----|----|---|---|

Unit – V

- | | | | | |
|--|----|----|---|---|
| 9. a) Describe the network level security issues in cloud computing | 10 | L1 | 5 | 1 |
| b) Describe the key benefits offered by multicloud in comparison to management of individual clouds. | 10 | L2 | 5 | 1 |
| 10. a) What are the limitations and advantages of cloud computing on business intelligence applications? | 10 | L2 | 5 | 1 |
| b) Describe the application security threats and risks in cloud computing. | 10 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
